March 15, 2025

Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)
National Science Foundation

Re: Response to the National Science's Foundation Request for Information

Respondent: Indeed

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Indeed welcomes the opportunity to submit comments to the National Science Foundation's (NSF) request for information (RFI) related to the "Development of an Artificial Intelligence Action Plan."

At Indeed, our mission is to help people get jobs. Indeed has over 595 million Job Seeker Profiles worldwide, across 60 countries, and through our products and services we enable fast connections to 24 million jobs every month. As a leading job search platform, Indeed sits at the intersection of labor supply and demand,[1] connecting millions of job seekers with employers across industries. This unique position provides deep insights into workforce needs, emerging skills,[2] and economic trends that shape the job market. By analyzing hiring patterns, skill gaps, and industry shifts, Indeed helps businesses and governments adapt to evolving workforce demands while empowering job seekers with the tools and resources to navigate changing career landscapes. As technology, particularly AI, reshapes work across sectors, Indeed's data-driven approach presents a clear understanding of how innovation impacts employment, skills development, and long-term economic growth.

[1] Hiring Lab's Global Jobs & Hiring Trends Reports for 2025
[2] The Transformational Opportunity of AI on ICT Jobs - AI Consortium

On Indeed, job seekers search for jobs, post resumes, and research companies, to find their next employment opportunity. Employers use Indeed to solve their recruitment and hiring challenges. Since Indeed first launched twenty years ago, AI has powered millions of connections between job seekers and employers on the platform. Thanks to AI, across the world, 23 job seekers are hired on Indeed every minute – someone gets hired on Indeed every three seconds. Our core mission is to help people get jobs, and help employers find great candidates - everywhere. AI is absolutely essential for large scale matching problems like employment. We welcome ongoing efforts to develop an actionable and scalable AI Action Plan, to establish clear expectations for the innovative and responsible development, deployment and use of AI.

AI and the Labor Market

AI is increasingly influencing the labor market, reshaping job roles, and creating new opportunities across various sectors. While AI automates certain tasks, it also augments human capabilities, leading to an important interplay between technology and employment. Human-centered skills[3] will remain essential in the workplace while AI and generative AI (GenAI) tools continue to evolve as their adoption in many industries continues to rise.

AI's Impact on Job Roles and Sectors. The share of US job postings mentioning GenAI, while still small overall,[4] has climbed rapidly from virtually nothing prior to late 2022, when the first generation of GenAI tools were widely introduced. And while the rapid growth of GenAI's presence in the workplace shows no signs of slowing, there are indicators that the nature of the growth is shifting, driven by a heightened focus on the implementation of GenAI tools, in addition to their ongoing development.

As of January 2024,[5] only about one job in 1,000 (0.1% of all jobs on Indeed) mentioned GenAI. By early 2025, that ratio had roughly tripled to 0.3%, a sign of rapidly accelerating employer interest in GenAI. But as rapid overall growth in GenAI mentions has continued, the types of jobs overtly noting GenAI are changing. In January 2024, the top 10 job titles mentioning GenAI were primarily for "makers" of the burgeoning technology, including machine learning engineers,

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[3] Hiring Lab Research Shows How Humans Will Remain an Essential Part of the Global Workforce as AI Evolves

[4] Rapid Growth in GenAI Job Postings: Expectations and Surprising Trends - Indeed Hiring Lab

[5] AI at Work: Rise of the GenAI Consultant - Indeed Hiring Lab

data scientists, and software architects/engineers. Fast forward a year, and those jobs are still prominent, but have been pushed down the list by another quickly growing role: management consultants. These professionals traditionally focus on helping businesses (their own or others) navigate change and improve processes — and, increasingly, to implement GenAI technologies.

Evolving Skill Requirements and Job Creation. The demand for AI expertise is transforming the technology job market, with nearly one in four tech job postings in the U.S. seeking candidates with AI skills. The information sector leads this trend, with 36% of its job postings being AI-related. This surge in demand highlights the importance of AI competencies for job seekers aiming to remain competitive.

The Indeed Hiring Lab assessed[6] the ability of GPT-4o, a GenAI model developed by OpenAI, to perform more than 2,800 job skills. Each skill was assessed across three main areas: The ability of GenAI to provide theoretical knowledge related to the skill; the ability of GenAI to solve problems using the skill; and GenAI's determination of the importance of physical presence in utilizing that skill. Even though GenAI's self-determed capabilities may carry significant overlap with human capabilities, current AI tools can't entirely replace a human worker's ability to seamlessly transition between subject matter expertise, problem-solving and physical execution in the workplace, however they still have enormous potential to boost productivity.

Overall, GenAI is generally strong at providing theoretical knowledge around many skills but less so at using skills to solve problems. And as long as a skill requires significant hands-on execution, the usefulness of GenAI may remain somewhat limited.

Even so, in jobs with a high share of skills that require hands-on execution, like nursing, GenAI could help with some repetitive tasks (like documentation) and allow workers to refocus on the core skills necessary in these roles. And in more stereotypical "office jobs" at which GenAI tools show more promise, including software development, the tools may potentially be able to offer significant knowledge and solve modest problems, emphasizing the importance of continued upskilling and ongoing learning for human workers.

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[6] AI at Work: Why GenAI Is More Likely To Support Workers Than Replace Them - Indeed Hiring Lab

Current and Future Perspectives on AI in the Workplace. According to an Indeed survey,[7] 87% of human resources and talent acquisition leaders and nearly 70% of job seekers are currently using AI systems and tools for everything from researching companies on the applicant side to drafting job descriptions on the hiring side. Another recent Indeed survey[8] found that hiring managers struggle with manually sourcing unresponsive candidates, while candidates wish companies would more thoroughly review their qualifications before contacting them about new opportunities. AI-enabled products and solutions are helping to make talent sourcing more efficient and impactful.

Building a Future-Ready AI Workforce

A strong workforce strategy is essential to ensuring that AI innovation supports economic mobility and job opportunities. The government should play a critical role in fostering AI adoption by setting the foundation for responsible innovation, ensuring AI advancements, and facilitating a regulatory environment that balances oversight with growth. By investing in AI research, supporting industry standards, and providing incentives for AI adoption, the government can accelerate the integration of AI tools and services across industries. Additionally, government-led AI adoption in public services—such as workforce development, talent attraction and human capital—can demonstrate AI's potential while de-risking implementation for private-sector enterprises.

A proactive government approach will not only strengthen national AI competitiveness but also drive widespread economic benefits through increased productivity, job creation, and technological resilience. A workforce strategy that prioritizes access to AI skills, economic opportunities, and talent retention will help ensure that AI advances economic opportunity for all. To foster a resilient and impactful AI workforce, the AI Action Plan should:

- Expand AI Workforce Development Initiatives. Promote federal support in STEM education, AI-focused vocational training, and research partnerships to cultivate a pipeline of highly skilled AI professionals.
- Enhance High-Skilled Employment Pathways. Streamline and strengthen programs to attract and retain top global AI talent, ensuring the U.S. remains a premier destination for AI expertise.

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[7] The Indeed Global AI Survey: Your Guide to the Future of Hiring
[8] Recruit Smarter, Not Harder, with the AI Efficiency of Indeed Smart Sourcing

- Support Industry-Led Training & Resource Access. Promote apprenticeships, public-private partnerships, and AI infrastructure accessibility for startups and small businesses to facilitate frequent and widespread AI development and adoption.
- Facilitate Workforce Upskilling. Introduce targeted opportunities for companies of all sizes, including small and mid-sized enterprises (SMEs), to invest in AI workforce training, enabling them to stay competitive in an AI-driven economy.
- Strengthening Public-Private Collaboration. Public-private partnerships are essential to maintaining U.S. leadership in AI. By fostering collaboration between industry, subject-matter experts, academia, and government, the U.S. can drive advancements in trustworthy AI, and workforce development and upskilling.

Fostering AI Innovation Through Smart, Adaptive Regulation
To sustain leadership in AI and expand economic opportunity, regulatory frameworks must be principles-based, forward-looking, and technology-neutral while balancing innovation and responsibility. A smart, adaptive approach will provide clarity, consistency, and global alignment, ensuring that AI-driven advancements benefit both businesses and workers. Indeed recommends the following principles for AI governance:

1. Align AI Regulation with Existing Frameworks to Prevent Redundancy
Before implementing new AI-specific regulations, policymakers should evaluate how existing laws—such as labor, civil rights, privacy, consumer protection, and cybersecurity regulations—already govern AI applications. A cohesive regulatory approach will:

- Minimize compliance burdens while ensuring responsible AI use.
- Reduce legal uncertainty by integrating AI oversight into existing governance structures.
- Support a federal privacy law to mitigate patchwork of laws and regulations, ensuring clarity for businesses and protecting consumer privacy while enabling AI-driven hiring innovations.

2. Prevent Regulatory Fragmentation & Promote Market Stability
A fragmented AI regulatory landscape—where compliance varies widely across jurisdictions—creates uncertainty, raises costs, and can disproportionately impact small businesses. To mitigate this:

- Establish a federal AI governance framework that interoperates with practical state laws to create consistency.
- Encourage ongoing regulatory and industry market practices to adapt policies to emerging AI developments rather than imposing rigid, one-size-fits-all mandates.
- Ensure AI regulation balances risk-based oversight with the flexibility needed to support AI-driven workforce platforms and job-matching technologies.
- A balanced AI innovation ecosystem that embraces both open and closed-source model development to foster competition, security, and broad industry adoption.

3. Implement Risk-Based, Scalable AI Compliance Standards

AI regulation should be proportionate to risk and application context, avoiding blanket compliance burdens that stifle innovation. A tiered regulatory approach should:

- Adjust oversight based on an AI system's risk level and intended use.
- Provide clear, tailored compliance obligations that reflect a company's role in AI development and deployment, ensuring fair applicability requirements across the ecosystem.
- Enable scalable compliance programs that support companies and applications of all sizes, without unnecessary barriers.

4. Strengthen Leadership in Global AI Governance

The U.S. must shape AI governance to ensure global competitiveness and facilitate cross-border AI adoption. Key actions include:

- Aligning AI regulations with relevant jurisdictions and international regulatory activity to streamline compliance for multinational businesses.
- Establishing U.S.-led AI competitiveness initiatives, with industry and foreign governments, to pursue practical, innovative and responsible AI policy frameworks and practices.
- Ensuring AI trade policies reduce foreign regulatory barriers, enabling U.S. businesses to compete fairly in AI-driven markets.

5. Advance Public-Private Collaboration to Expand Workforce Opportunity

Industry, academia, and government must work together to ensure AI adoption supports safe use, economic growth and job creation. Key focus areas:

- Trustworthy AI in hiring – Collaborate on standards for fairness, explainability, and ethical AI use in workforce applications.
- AI accessibility for businesses – Ensure small businesses and startups can leverage AI to enhance hiring and workforce efficiency.
- Investment in AI infrastructure – Support development of new computing resources, and efficient AI advancements to drive long-term growth and innovation.

Advancing AI to Expand Economic Opportunity
Indeed greatly appreciates NSF's leadership in developing an AI Action Plan to foster AI-driven innovation that benefits workers, job seekers, and businesses. As AI reshapes hiring and the labor market, policies must balance risk and opportunity to ensure effective and safe participation in AI-driven growth. Overly rigid regulations could stifle AI tools that connect people to jobs and enhance hiring efficiency. Indeed remains committed to policies that empower workers and employers and looks forward to continued collaboration in building an AI ecosystem that drives job growth, productivity, and economic opportunity.

Respectfully,

Matthew Jensen
Head of Government Relations and Public Policy, Indeed